

Spam email detection using BERT and GPT models

Qinyuan Qi

Liu ID: qinqi464

Abstract

Spam email detection remains a critical challenge in cybersecurity, as malicious emails continue to evolve in sophistication. Traditional methods often rely on handcrafted features or shallow machine learning models, which struggle to capture the complex semantic and contextual patterns in modern spam emails. In this paper, we try to implement a spam email detection system using the GPT model while using BERT as the base model to find out the potential performance improvement of the GPT model.

After comparing the performance of the two models, we found that the BERT model can greatly improve the performance of the spam detection rate, and for GPT model, training time increases a lot and result is not good compared to the BERT model.

1 Introduction

With the rapid development of the Internet, e-mail has become a fast and convenient methods of communication and has been widely used in our daily lives.

However, the popularity of e-mail has also brought about a problem that cannot be ignored: spam. Unscrupulous businessmen and criminals have seized this communication medium of e-mail and started to send spam with commercial advertisements, and malicious or illegal files to users on a large scale(Kumar et al., 2020). Spam not only occupies network resources and causes network congestion, but also affects people's normal work and life.

Therefore, how to quickly and accurately check whether an email is spam or not becomes a major topic worthy of research. It can protect people's property and privacy security and also improve people's work and life efficiency.

In this paper, we will try to solve spam text detection, using the latest TREC 2007 spam email

dataset(Cormack, 2007) to train our model. We will use the BERT model as the baseline model and compare it with the GPT model to find out which model is more suitable for spam email detection.

2 Theory

2.1 Transformer & Attention

The transformer architecture(Vaswani et al., 2023) revolutionized natural language processing by replacing traditional recurrent and convolutional layers with a mechanism based entirely on self-attention. This innovation enables the model to process entire text sequences in parallel, capturing long-range dependencies and contextual relationships more effectively than previous architectures. Figure 1 shows the architecture of the transformer model, it consists two parts, encoder and decoder.

Building upon the foundational Transformer architecture, BERT (Bidirectional Encoder Representations from Transformers) and GPT (Generative Pre-trained Transformer) have emerged as two of the most influential models in the field of natural language processing (NLP).

BERT(Koroteev, 2021) is designed as a bidirectional deep learning model that captures contextual information from both the left and right sides of a given word in a sentence. This bi-directionality enables BERT to gain a deeper understanding of language structure, making it particularly suitable for tasks such as question answering, named entity recognition, and sentence classification. BERT is trained using the Masked Language Model (MLM) objective (Koroteev, 2021), where a subset of words in a sentence are randomly masked and predicted based on the surrounding context, and the Next Sentence Prediction (NSP) task, which helps the model learn relationships between sentences.

In contrast, GPT and its subsequent iterations (GPT-2, GPT-3, and GPT-4) adhere to an autoregressive modelling approach.(Zhao et al., 2025).

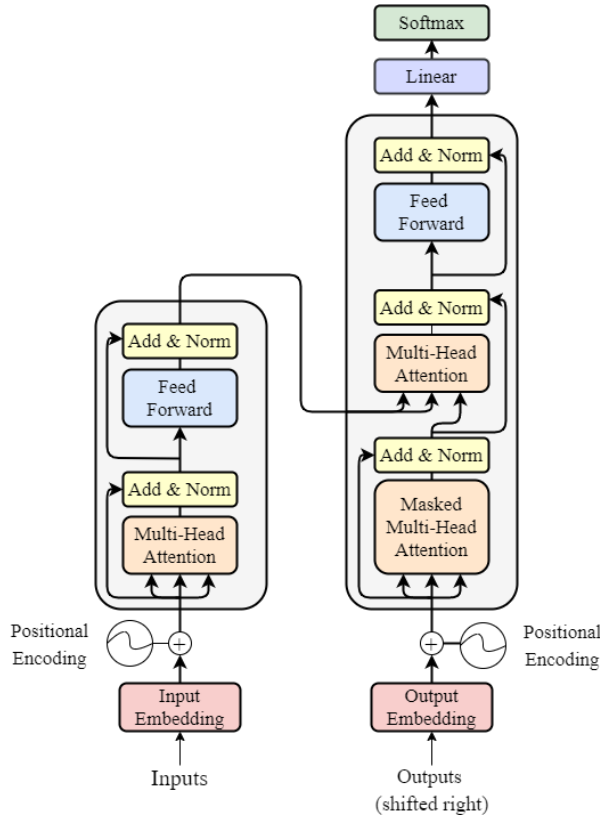


Figure 1: Transformer architecture

GPT generates text sequentially by predicting the next token based on the previous tokens in a left-to-right manner. Unlike BERT, which uses bidirectional attention, GPT uses unidirectional attention, limiting its focus to the previously generated tokens in the sequence. The model is pre-trained using the Causal Language Modeling (CLM) objective (Zhao et al., 2025), where it learns to predict the next word given a context, making it very effective in text generation, dialogue systems, and creative content generation.

Although both BERT and GPT leverage the self-attention mechanism inherent in Transformers, they employ different architectural components and training methods, resulting in unique capabilities and applications.

3 Data

This paper uses the trec07 spam corpus (Cormack, 2007) as the experimental dataset downloaded from Kaggle, and it is a public dataset provided by the International Conference on Text Retrieval. The trec07 dataset includes 50,199 spam emails and 25,220 ham emails. This paper randomly chooses 80% as training data and uses the remaining 20% as test data.

The data columns are as follows:

- **label:** Email label, 1 for spam / 0 for ham.
- **origin:** Email content, containing all the text in a email.

4 Method

4.1 Data Preprocessing

The data preprocessing steps used in this paper including the following steps:

- Replace the abbreviations with their normal form.
- Remove all the stop words.
- Lemmatize the words with its normal form.

We will use functions provided by the NLTK library to remove the stop words and lemmatize the words. Meanwhile, we also use the regular expressions library to replace the abbreviations with their normal form to make the data more clean and easy to process.

After the data preprocessing, the data will be tokenized and converted into a format that can be used by the model.

Then the processed data will be split into training (80%) and testing data (20%).

4.2 Email spam text classification process

The Email spam text classification process is shown in figure 2 below. The process includes the following steps:

- Preprocessing the data
- Tokenization
- Model building and training
- Spam mail classification

To check the performance of the models, the models will also be evaluated using the test data mentioned above.

4.3 BERT model

The BERT model used here is a pre-trained model sentence-transformers/bert-base-nli-mean-tokens (Reimers and Gurevych, 2019) which can be downloaded from the huggingface model hub. It contains 2 linear layers with 256 hidden units, dropout rate set to 0.2, and use RELU function as activation function. Loss function used in BERT model is Binary Cross Entropy Loss, and optimizer used is Adam optimizer with default learning rate.

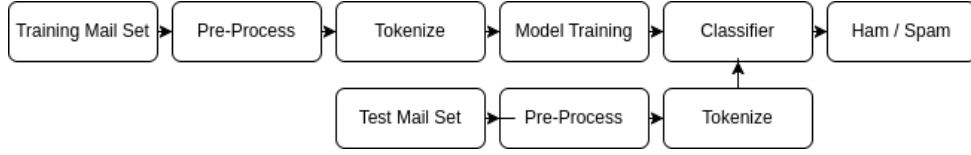


Figure 2: Mail classification process

4.4 GPT model

GPT model, on the other hand, use the pre-trained model gpt2 provided OpenAI(Radford et al., 2019), and it share the same SpamClassifier class with the BERT model. However, GPT model are send to GPU instead of CPU, and required input data are modified to fit the GPT model accordingly. Loss function used in GPT model is Cross Entropy Loss instead of Binary Cross Entropy Loss, and optimizer used is Adam optimizer with default learning rate.

GPT model contains 2 linear layers with 256 hidden units, dropout rate set to 0.2, and use RELU function as activation function.

4.5 Evaluation

This article uses following four values as evaluation indicators. 1.Accuracy. 2.Precision. 3.Recall. 4.F1-score. Accuracy represents the spam detection rate, which is the accuracy of the model's classification results, Precision represents the accuracy of the classifier classification, recall represents the spam detection rate and F1 value is a combination of the precision and recall of the classifier, which is the harmonic average of the two, and it is a value between 0 and 1.

5 Results

5.1 Experiment Environment

The experiments in this paper are conducted on Google Colab(Google) with 1 CPU, 12GB RAM, and an NVIDIA GeForce T4 graphics card. The main libraries used in this experiment includes nltk, pytorch, scikit-learn, transformers and matplotlib libraries. For the code related to this experiment, please refer to the following link.¹

5.2 Experiment Results

In this experiment, we employed the BERT model for training and monitored the loss and accuracy, as illustrated in Figure 3. The plotted results indicate that the training loss consistently decreased



Figure 3: BERT model Loss and Accuracy

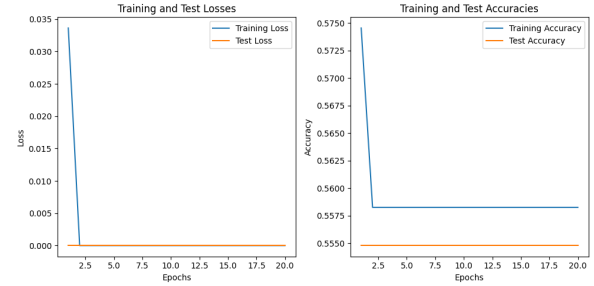


Figure 4: GPT model Loss and Accuracy

from 0.14 to 0.02. Meanwhile, the validation loss exhibited a decreasing trend from approximately 0.1 to 0.07, albeit with fluctuations. However, after 12 epochs, an increase in the validation loss was observed, suggesting potential overfitting.

For the Accuracy, training accuracy increased from 0.94 to 0.99, while the validation accuracy increased from around 0.964, and reached around 0.98, but still with some fluctuations when epoches increasing. Over all, the BERT model archive a pretty good performance in this experiment.

For the GPT2 model, we notice that the accuracy is pretty low(around 0.55) compare to the BERT model(around 0.98), regarding the loss value, it is drop to 0 after 2 epoches.

We list the final four values of the BERT and GPT2 models in Table 1.

6 Discussion

From the previous result, we notice that BERT model perform much better than the GPT model in

¹Source Code:https://github.com/qyfly/text_mining_project

Model	BERT	GPT2
Accuracy	0.97987	0.44522
Precision	0.97928	0
Recall	0.98455	0
F1-score	0.97987	0.27431

Table 1: BERT and GPT model evaluation

the spam email detection task. This is because of the following reasons.

1. Fundamental architectural differences between the two models.

We know that BERT Uses Bidirectional Context, meaning it considers both the left and right context when encoding a word. This allows BERT to capture rich contextual information about the entire email, making it well-suited for classification tasks where understanding the whole text is crucial.

GPT-2 is an autoregressive model, meaning it generates text from left to right, only considering past tokens (unidirectional). This limits its ability to understand the full context of an email, especially when key spam indicators appear later in the text.

So BERT is designed for Classification, GPT-2 is designed for Text Generation.

2. Token Representation Differences

BERT processes the entire input at once, creating a global representation of the text.

GPT-2 processes tokens sequentially, which may result in it losing key contextual clues by the time it reaches the end of an email.

Regarding the limitations of this work, we can point out the following:

1. Limited Dataset

We only use the text data instead of html mail which may contains more multimedia information, such as picture and video. So the model we use may not be able to detect spam emails with multimedia information.

2. Limited Models

We only use the BERT and GPT model to detect spam emails, and there are many other models that can be used to detect spam emails.

3. Limited Evaluation Metrics

We only use accuracy, precision, recall, and F1-score as evaluation indicators, and there are many other evaluation indicators that can be used to evaluate the model. for example, ROC curve, etc.

About the future work and the limitations mentioned above, we need use more advanced models

to detect spam emails, such as RoBERTa, etc. to identify those spam emails with multimedia information.

7 Conclusion

In this study, we aimed to compare the performance of BERT and GPT-2 in the task of spam email classification. Given their distinct architectures—BERT being a bidirectional transformer and GPT-2 being an autoregressive model—we hypothesized that their ability to classify spam emails would differ significantly.

7.1 Key Findings

After training and evaluating both models, we found that BERT significantly outperformed GPT-2 in classifying spam emails. The primary reasons for this performance gap stem from the fundamental differences in how these models process text:

- **Bidirectional vs. Unidirectional Processing:** BERT's ability to consider both past and future tokens in a text sequence allowed it to develop a richer contextual understanding. In contrast, GPT-2's left-to-right processing limited its ability to capture dependencies that extend beyond previous words, leading to a weaker grasp of the full email content.
- **Optimization for Classification:** BERT's pre-training objectives, including masked language modeling and next-sentence prediction, made it inherently better suited for classification tasks. Its [CLS] token serves as an effective representation for the entire input text, simplifying fine-tuning for spam classification. GPT-2, primarily designed for text generation, lacks a specialized mechanism for holistic text representation, making it less effective for classification.
- **Token Representation and Attention Mechanisms:** BERT processes entire email texts simultaneously, enabling it to establish long-range dependencies across the input. GPT-2, by contrast, processes tokens sequentially, which may cause it to overlook crucial spam indicators located later in the text.

7.2 Extent to Which the Problem Was Solved

Beyond just the comparative performance of BERT and GPT-2, we gained several insights from this study:

- **Model Selection Matters:** Choosing the right transformer model based on the specific NLP task is crucial. While GPT-2 excels at text generation, BERT is more effective for classification.
- **Pretraining Objectives Influence Performance:** The way a model is pretrained significantly impacts its downstream performance. Classification tasks benefit from bidirectional training objectives.
- **Interpretability and Efficiency:** BERT's structured output makes it easier to interpret classification decisions compared to GPT-2's token-by-token processing, which lacks a global representation of the input.

In this study, we aim to compare the performance of BERT and GPT-2 on the task of spam classification.

Since they have different architectures, BERT is a bidirectional Transformer, while GPT-2 is an autoregressive model. Considering their good performance of GPT model for most of the tasks, we hypothesize GPT will perform better than BERT in spam classification.

After training and evaluating both models, we find that BERT significantly outperforms GPT-2 on spam classification.

This performance gap stems primarily from fundamental differences in how they process text:

- **Bidirectional vs. unidirectional processing:** BERT considers both preceding and following words in a text sequence, building a richer contextual understanding. In contrast, GPT-2 uses a left-to-right approach and can only rely on previous words, limiting its ability to capture long-range dependencies, making it less able to understand the entire email content.
- **Difference in optimization objectives:** BERT's pre-training objectives include Masked Language Modeling and Next Sentence Prediction, making it more suitable for classification tasks. The token can be used as an effective representation of the entire input text, thus simplifying the fine-tuning process for spam classification. In contrast, GPT-2 is mainly used for text generation and does not have a dedicated overall text representation mechanism, so it performs poorly on classification tasks.

- **Token representation and attention mechanism:** BERT processes the entire email text and is able to establish long-distance dependencies in the input sequence. GPT-2 processes tokens sequentially, which may ignore important spam features in the second half of the email.

In addition to comparing the performance of BERT and GPT-2, we also gained several important insights from this study:

- **Model selection is crucial:** Choosing the right Transformer model for a specific natural language processing (NLP) task is crucial. GPT-2 performs well in text generation, while BERT is more efficient in classification tasks.
- **Pre-training objectives affect model performance:** The pre-training method of the model can significantly affect the performance of its downstream tasks. Classification tasks often benefit from bidirectional training objectives.
- **Interpretability and efficiency:** BERT's structured output makes its classification decisions more explainable, while GPT-2's word-by-word prediction method lacks global text representation, making its classification ability weaker.

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